SOMEBODY QUIETLY RELEASED A WHOLE VIRTUAL LAB

The Virtual Neuro Lab

Experiments on a complete fruit-fly brain: 139,255 neurons, wired from the FlyWire map.

Romana · Powered by Meta’s Muse



WHY A FLY?

Her whole brain has been mapped, cell by cell

139,255

neurons in the adult fruit-fly brain, mapped with their connections by the FlyWire project. One of the most complex brains ever mapped, neuron by neuron.

~75%

of the genes linked to human diseases have a fly version. What we learn in her brain can point to ours.

THE VIVARIUM

Meet Fly: a real body, a real brain



Her body comes from a micro-CT scan of a real fruit fly. Her brain is the FlyWire wiring. Choose what she senses:

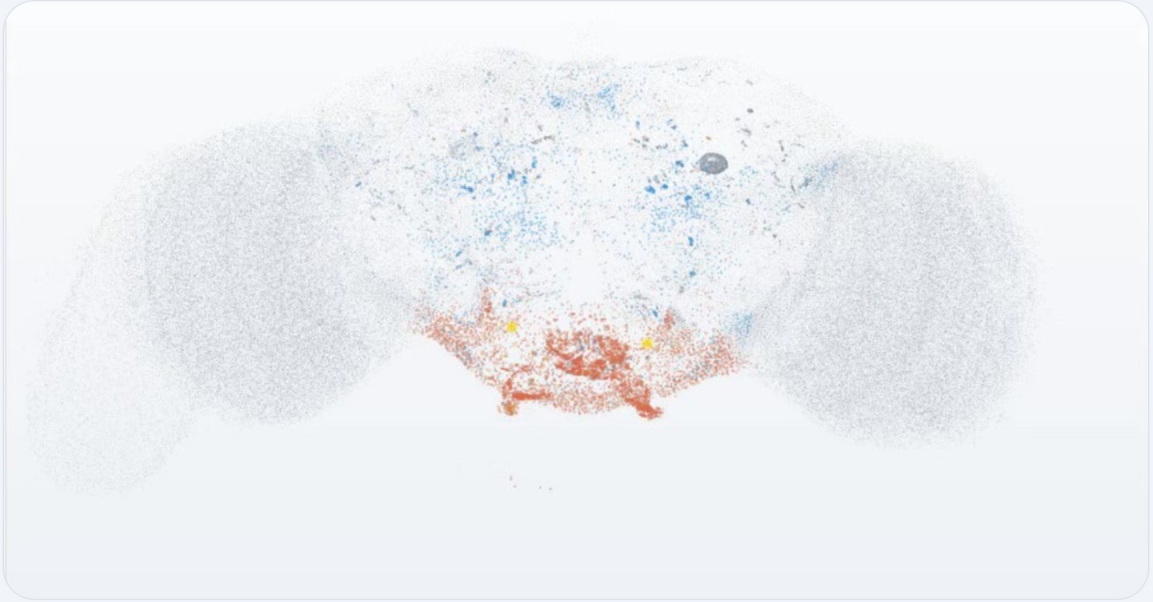
Sugar → tongue out to feed

Sugar + bitter → she refuses

Touch her antennae → she grooms

INSIDE HER BRAIN

Watch her brain decide



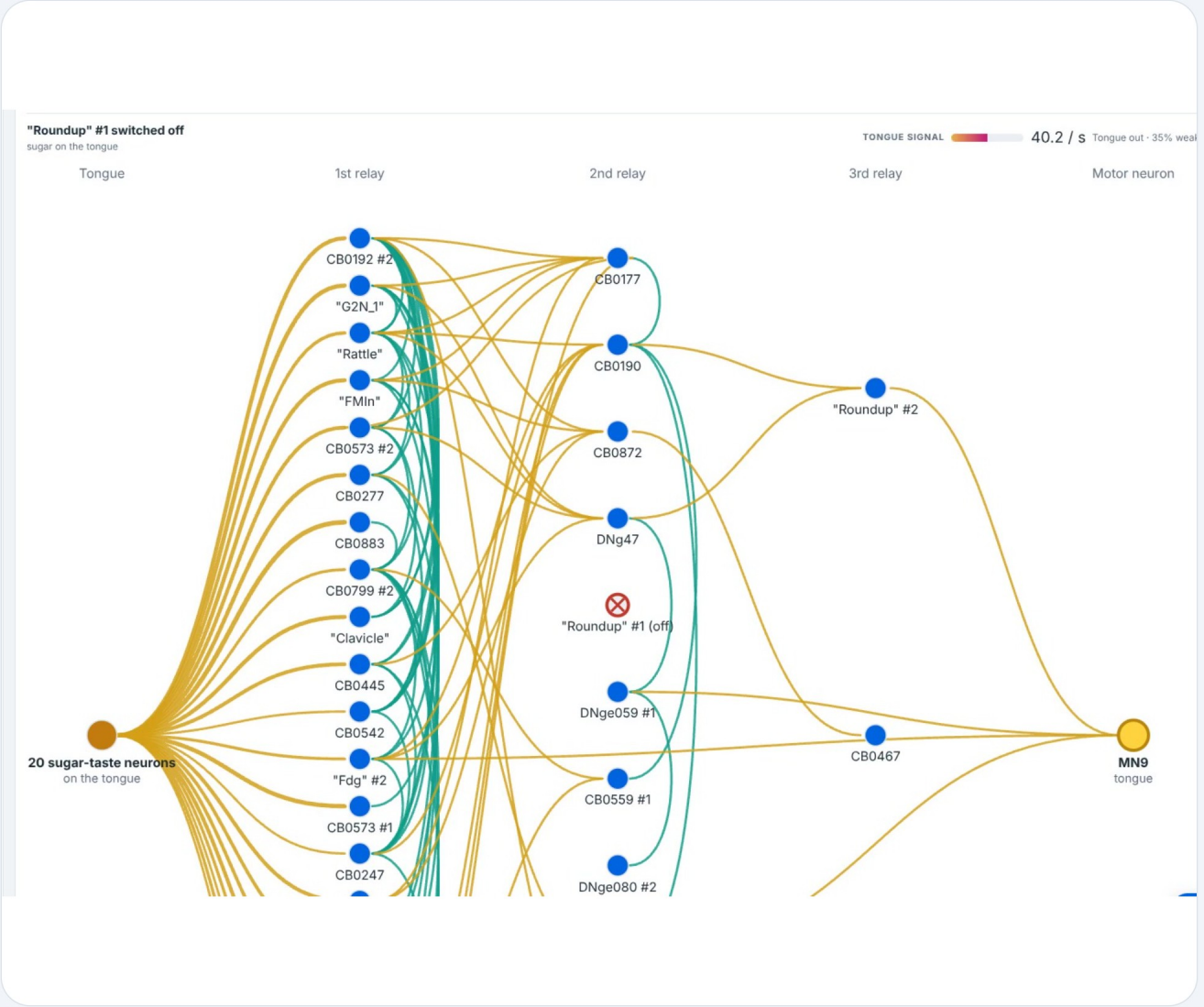
1. **Signal in.** We switch on 20 sugar-taste neurons. Nothing else.
2. **Processing.** The brain’s own wiring carries it: 385 neurons fire, mostly in the taste and feeding centre.
3. **Decision.** MN9, the tongue neuron, fires 65 times a second. Over 10, so the tongue comes out.

NEURON SILENCING SCREEN

Switch off one neuron. See what breaks.

With “Roundup” #1 switched off, the sugar signal finds other routes, but the tongue neuron fires **35% less**.

The model predicts the fly needs this neuron to eat.



ALREADY FINDING THINGS

286 neurons switched off, one at a time

286

neurons screened for eating and drinking

8

the fly seems to need. 2 of them agree with real-fly experiments

3

never tested in a living fly, as far as we can find: new predictions

The three new candidates: **CB0051**, **CB0192 #2** and **CB0501**, each predicted to be needed for drinking.

WHY TRUST IT

10 / 15

results already known from real flies, matched by the untouched brain.

It passed its calibration test first: sugar makes the tongue come out, bitter stops it.

Nothing was adjusted to pass. Every pass mark was written down before the run.

The misses stay on the page, so you can see where it’s weaker.

POWERED BY META'S MUSE

Ask Muse anything about her

YOU ASK

"Has anyone tested CB0051 in a real fly?"

MUSE LOOKS

In the lab's own records first, then on the web

MUSE ANSWERS

In plain words, with every source shown

Muse sits on every page of the lab, so anyone can ask what a result means, without a biology degree.

WHY IT MATTERS

Two reasons

1

The brain is still mostly a mystery

A whole brain you can switch on and off, cell by cell, shows how its parts work together.

2

Drug trials

About 9 in 10 drugs that reach human trials fail. A predictive lab could rule out bad ideas sooner, cutting time, money, and risk to lives.

WHAT'S NEXT

From one fly toward drug discovery

NOW

A partner fly lab

to test our 3 new predictions in living flies

IN THE LAB

Parkinson's

which neurons lose their dopamine as dopamine cells die

NEXT STEP

Docking

does a drug molecule actually fit its target?

LONG TERM

A virtual patient

who tries the drug first, so nobody else has to

NOW YOU KNOW ABOUT IT

The Virtual Neuro Lab

Powered by Meta’s Muse, for humans. And this is only the first fly.

github.com/ummeromana21-creator/TheVirtualNeuroLab

